

Chapter 3 – Rigid-Body Kinetics

3.1 Newton–Euler Equations of Motion about the CG

3.2 Newton–Euler Equations of Motion about the CO

3.3 Rigid-Body Equations of Motion

To derive the marine craft equations of motion, it is necessary to study of the **motion of rigid bodies**, **hydrodynamics**, and **hydrostatics**.

The overall goal of Chapter 3 is to show that the rigid-body equations of motion can be expressed in matrix-vector form about the CO according to the following:

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

$\mathbf{M}_{RB}$ Rigid-body mass matrix

$\mathbf{C}_{RB}$ Rigid-body Coriolis and centripetal matrix due to the rotation of {b} about {n}

$\boldsymbol{\nu} = [u, v, w, p, q, r]^T$ generalized velocity expressed in {b}

$\boldsymbol{\tau}_{RB} = [X, Y, Z, K, M, N]^T$ generalized force expressed in {b}

CO - coordinate origin o_b of {b}

CG - center of gravity relative to the CO, located at $\mathbf{r}_{bg}^b = [x_g, y_g, z_g]^T$

CO is the body-fixed coordinate origin of the GNC system, and it is defined to meet specific control objectives. As such, the craft's **CO** can be stabilized or programmed to follow a time-varying trajectory in 2-D or 3-D space.

第3章 – 刚体动力学

3.1 关于质心的牛顿-欧拉运动方程 3.2 关于质心系的牛顿-欧拉运动方程 3.3 刚体运动方程

为了推导海上船舶的运动方程，有必要研究刚体运动、流体力学和静水力学。

第三章的总体目标是展示刚体运动方程可以根据以下内容以矩阵-向量形式围绕质心表示：

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

$\mathbf{M}_{RB}$ 刚体质量矩阵 $\mathbf{C}_{RB}$ 由于 $\{b\}$ 关于 $\{n\}$ 的旋转引起的刚体科里奥利和向心矩阵 $\mathbf{v} = [u, v, w, p, q, r]^T$ 在 $\{b\}$ 中表示的广义速度 $\boldsymbol{\tau}_{RB} = [X, Y, Z, K, M, N]^T$ 在 $\{b\}$ 中表示的广义力

CO是GNC系统的机体固定坐标原点，它的定义是为了满足特定的控制目标。因此，飞行器的CO可以被稳定或编程以在二维或三维空间中跟随时间变化的轨迹。

CO - coordinate origin o_b of $\{b\}$

CG - center of gravity relative to the CO, located at $\mathbf{r}_{bg}^b = [x_g, y_g, z_g]^T$

Chapter Goals

- Understand that Newton's 2nd law and its generalization to the **Newton-Euler equations of motion is formulated in an "approximative" inertial frame** usually chosen as the tangent plane NED.
- Understand why we get a $\mathbf{C}_{RB}$ matrix when transforming the equations of motion to a BODY-fixed rotating reference frame instead of NED.
- Be able to write down and simulate the **rigid-body equation of motion about the CG** for a vehicle moving in 6 DOFs.
- Know how to **transform** the rigid-body equation of motion **to other reference points** such as the CO. This involves using the $\mathbf{H}$ -matrix (system transformation matrix) defined in Appendix C.
- Understand the:
 - 6 x 6 rigid-body matrix $\mathbf{M}_{RB}$
 - 6 x 6 Coriolis and centripetal matrix $\mathbf{C}_{RB}$ and how it is computed from $\mathbf{M}_{RB}$
 - Parallel-axes theorem and its application to moments of inertia

章节目标

- 要理解牛顿第二定律及其推广到牛顿-欧拉运动方程，是在一个“近似的”惯性参考系中表述的，这个参考系通常选择为切平面NED。
- 理解为什么当将运动方程转换到机体固定的旋转参考系而不是NED时，我们会得到一个 $\mathbf{C}_{RB}$ 矩阵。
- 能够写出并模拟车辆在六自由度运动中关于质心的刚体运动方程。
- 了解如何将刚体运动方程转换到其他参考点，例如质心。这涉及使用附录C中定义的 $\mathbf{H}$ 矩阵（系统转换矩阵）。
- 理解：- 6 x 6 刚体矩阵 $\mathbf{M}_{RB}$ - 6 x 6 科里奥利和向心矩阵 $\mathbf{C}_{RB}$ 以及它如何从 $\mathbf{M}_{RB}$ 计算得出 - 平行轴定理及其在转动惯量中的应用

Chapter 3 – Rigid-Body Kinetics

The equations of motion will be represented in two body-fixed reference points

- 1) Center of gravity (CG), subscript g
- 2) Origin CO of $\{b\}$, subscript b

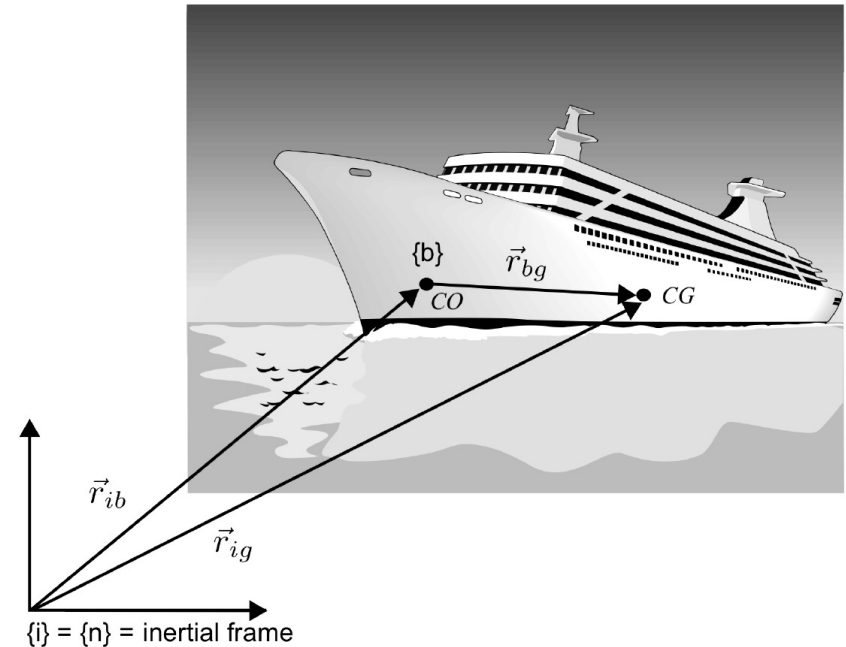
These points coincides if the vector $\vec{r}_{bg}^b = \mathbf{0}$

Time differentiation of a vector in a moving reference frame $\{b\}$ satisfies

$$\frac{{}^i d}{dt} \vec{a} = \frac{{}^b d}{dt} \vec{a} + \vec{\omega}_{ib} \times \vec{a}$$

Time differentiation in $\{b\}$ is denoted as

$$\dot{\vec{a}} := \frac{{}^b d}{dt} \vec{a}$$



第3章 – 刚体动力学

运动方程将以两个固定于物体的参考点表示

1) 重心 (CG), 下标 g 2) $\{b\}$ 的原点
CO, 下标 b

如果向量, 这些点重合

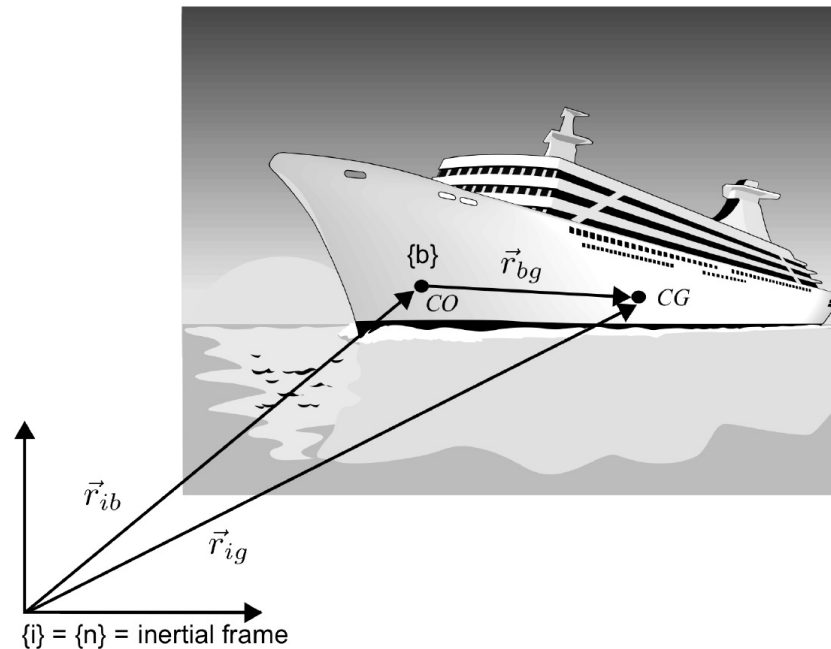
$$\mathbf{r}_{bg}^b = \mathbf{0}$$

在运动参考系 $\{b\}$ 中, 向量的时间微分
满足

$$\frac{{}^i d}{dt} \vec{a} = \frac{{}^b d}{dt} \vec{a} + \vec{\omega}_{ib} \times \vec{a}$$

$\{b\}$ 中的时间微分表示为

$$\dot{\vec{a}} := \frac{{}^b d}{dt} \vec{a}$$



3.1 Newton-Euler Equations of Motion about the CG

Coordinate-free vector: A vector $\vec{v}_{nb}$, velocity of {b} with respect to {n}, is defined by its magnitude and direction but without reference to a coordinate frame.

Coordinate vector: A vector $\vec{v}_{nb}$ decomposed in the inertial reference frame is denoted by

Newton-Euler Formulation

Newton's Second Law relates mass m , acceleration $\dot{\vec{v}}_{ig}$ and force $\vec{f}_g$ according to

$$m\dot{\vec{v}}_{ig} = \vec{f}_g$$

where the subscript g denotes the center of gravity (CG).

Euler's First and Second Axioms

Euler suggested to express Newton's Second Law in terms of conservation of both linear momentum $\vec{p}_g$ and angular momentum $\vec{h}_g$ according to:

$$\begin{aligned} \frac{d}{dt}\vec{p}_g &= \vec{f}_g & \vec{p}_g &= m\vec{v}_{ig} \\ \frac{d}{dt}\vec{h}_g &= \vec{m}_g & \vec{h}_g &= I_g\vec{\omega}_{ig} \end{aligned}$$

$\vec{f}_g$ and $\vec{m}_g$ are forces/moments about the body's CG
 $\vec{\omega}_{ib}$ is the angular velocity of frame b relative frame i
 I_g is the inertia dyadic about the body's CG



Isaac Newton (1642-1726)
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Leonhard Euler (1707-1783)
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3.1 关于质心的牛顿-欧拉运动方程

无坐标矢量：一个矢量 $\vec{v}_{nb}$ ， $\{\mathbf{b}\}$ 相对于 $\{\mathbf{n}\}$ 的速度，通过其大小和方向定义，但不参考坐标系。坐标矢量：在惯性参考系中分解的矢量 $\vec{v}_{nb}$ 表示为

牛顿-欧拉公式

牛顿第二定律根据质量 m 、加速度 $\dot{\vec{v}}_{ig}$ 和力 $\vec{f}_g$ 的关系

$$m\dot{\vec{v}}_{ig} = \vec{f}_g$$

下标 g 表示重心 (CG)。

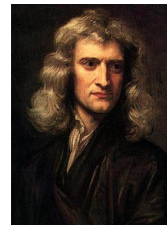
欧拉的第一和第二公理

欧拉建议用守恒的方式表达牛顿第二定律

根据：线性动量和角动量 $\vec{h}_g$

$$\begin{aligned} \frac{d}{dt} \vec{p}_g &= \vec{f}_g & \vec{p}_g &= m\vec{v}_{ig} \\ \frac{d}{dt} \vec{h}_g &= \vec{m}_g & \vec{h}_g &= I_g \vec{\omega}_{ig} \end{aligned}$$

$\vec{f}_g$ 并且作用在物体质心上的力/力矩是相对于参考系 i 的参考系 b 的角速度 I_g 是物体质心的惯性二重项



艾萨克·牛顿 (1642-1726)
3) 维基共享资源



莱昂哈德·欧拉 (1707-1783)
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3.1 Translational Motion about the CG

When deriving the equations of motion, it will be assumed that:

- (1) The vessel is rigid
- (2) The NED frame is inertial—that is, $\{n\} \approx \{i\}$

The first assumption eliminates the consideration of forces acting between individual elements of mass while the second eliminates forces due to the Earth's motion relative to a star-fixed inertial reference system such that

$$\begin{aligned}\vec{v}_{ig} &\approx \vec{v}_{ng} \\ \vec{\omega}_{ig} &= \vec{\omega}_{ib} \approx \vec{\omega}_{nb}\end{aligned}$$

For guidance and navigation applications in space, it is usual to use a star-fixed reference frame or a reference frame rotating with the Earth. Marine craft are, on the other hand, usually related to the NED reference frame. This is a reasonable assumption since forces on a marine craft due to the Earth's rotation

$$\omega_{ie} = 7.2921 \times 10^{-5} \text{ rad/s}$$

are quite small compared to the hydrodynamic forces.

3.1 关于重心的平动运动

在推导运动方程时，将假设：

- (1) 该容器是刚性的
- (2) NED 框架是惯性的——也就是说， $\{n\} \approx \{i\}$

第一个假设排除了考虑作用在各个质量元素之间的力，而第二个假设排除了由于地球相对于恒星固定惯性参考系运动而产生的力，因此

$$\begin{aligned}\vec{v}_{ig} &\approx \vec{v}_{ng} \\ \vec{\omega}_{ig} &= \vec{\omega}_{ib} \approx \vec{\omega}_{nb}\end{aligned}$$

在空间中的导航和引导应用中，通常使用恒星固定参考系或随地球旋转的参考系。另一方面，海上船舶通常与NED参考系相关。这是一个合理的假设，因为地球自转对海上船舶的作用力

$$\omega_{ie} = 7.2921 \times 10^{-5} \text{ rad/s}$$

相比于流体动力，尺寸相当小。

3.1 Translational Motion about the CG

$$\vec{r}_{ng} = \vec{r}_{nb} + \vec{r}_{bg} \quad \leftarrow \text{{n} is inertial} \quad \vec{r}_{ig} = \vec{r}_{ib} + \vec{r}_{bg}$$

Time differentiation of $\vec{r}_{ng}$ in a moving reference frame {b} gives

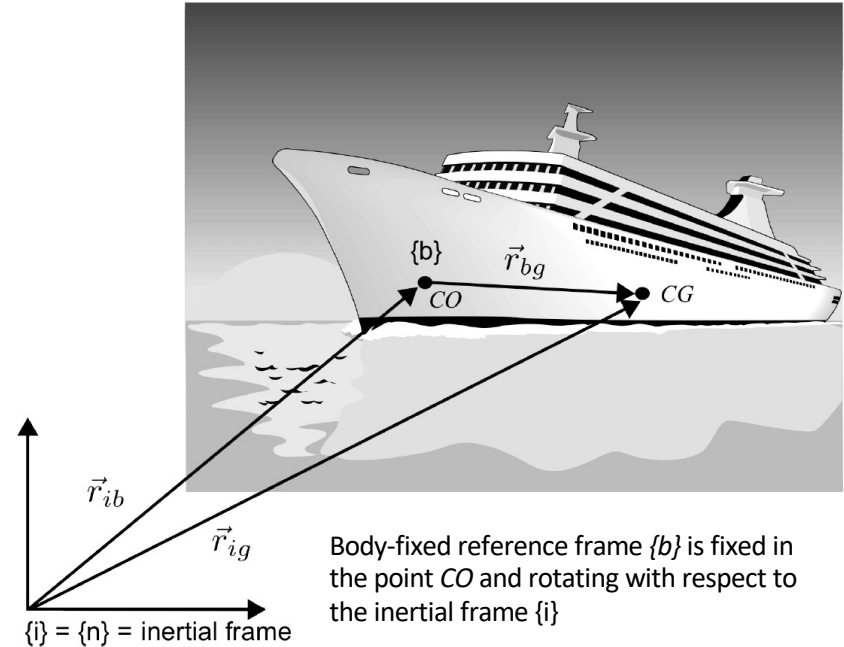
$$\vec{v}_{ng} = \vec{v}_{nb} + \left(\frac{b d}{dt} \vec{r}_{bg} + \vec{\omega}_{nb} \times \vec{r}_{bg} \right)$$

For a rigid body, the CG satisfies

$$\frac{b d}{dt} \vec{r}_{bg} = \vec{0}$$

$$\vec{v}_{ng} = \vec{v}_{nb} + \vec{\omega}_{nb} \times \vec{r}_{bg}$$

$$\begin{aligned} \vec{f}_g &= \frac{i d}{dt} (m \vec{v}_{ig}) \\ &= \frac{i d}{dt} (m \vec{v}_{ng}) \\ &= \frac{b d}{dt} (m \vec{v}_{ng}) + m \vec{\omega}_{nb} \times \vec{v}_{ng} \\ &= m (\dot{\vec{v}}_{ng} + \vec{\omega}_{nb} \times \vec{v}_{ng}) \end{aligned}$$



Translational Motion about the CG Expressed in {b}

$$m [\dot{\vec{v}}_{ng}^b + \mathbf{S}(\boldsymbol{\omega}_{nb}^b) \vec{v}_{ng}^b] = \vec{f}_g^b$$

3.1 关于重心的平动运动

$$\vec{r}_{ng} = \vec{r}_{nb} + \vec{r}_{bg} \quad \leftarrow \quad \vec{r}_{ig} = \vec{r}_{ib} + \vec{r}_{bg}$$

{n} is inertial

在运动参考系 {b} 中的时间微分得到

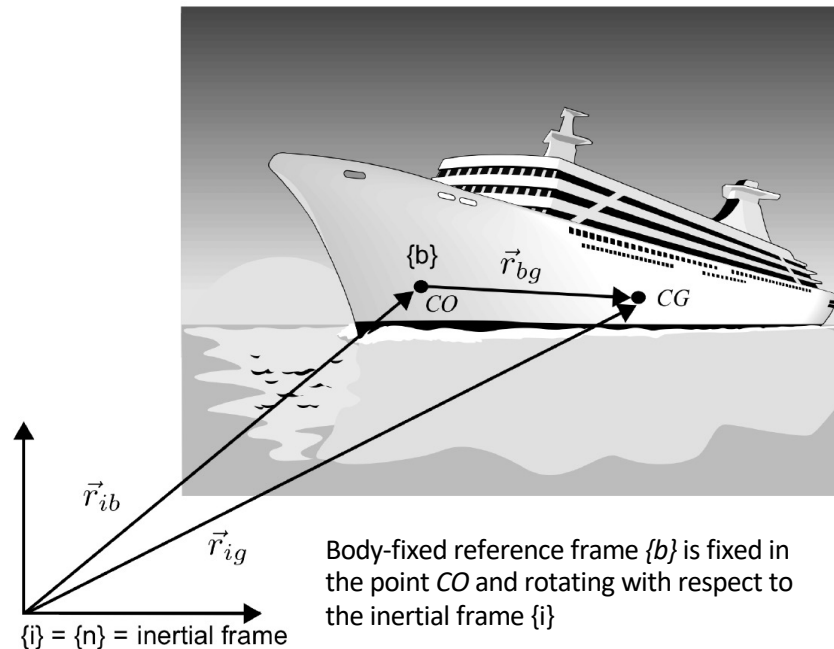
$$\vec{v}_{ng} = \vec{v}_{nb} + \left(\frac{b}{dt} \vec{r}_{bg} + \vec{\omega}_{nb} \times \vec{r}_{bg} \right)$$

对于刚体，质心满足

$$\frac{b}{dt} \vec{r}_{bg} = \vec{0}$$

$$\vec{v}_{ng} = \vec{v}_{nb} + \vec{\omega}_{nb} \times \vec{r}_{bg}$$

$$\begin{aligned} \vec{f}_g &= \frac{i}{dt} (m \vec{v}_{ig}) \\ &= \frac{i}{dt} (m \vec{v}_{ng}) \\ &= \frac{b}{dt} (m \vec{v}_{ng}) + m \vec{\omega}_{nb} \times \vec{v}_{ng} \\ &= m (\dot{\vec{v}}_{ng} + \vec{\omega}_{nb} \times \vec{v}_{ng}) \end{aligned}$$



Translational Motion about the CG Expressed in {b}

$$m[\dot{\vec{v}}_{ng}^b + \mathbf{S}(\boldsymbol{\omega}_{nb}^b) \vec{v}_{ng}^b] = \vec{f}_g^b$$

3.1 Rotational Motion about the CG

The derivation starts with the Euler's 2nd axiom

$$\begin{aligned}\vec{m}_g &= \frac{d}{dt}(I_g \vec{\omega}_{ib}) \\ &= \frac{d}{dt}(I_g \vec{\omega}_{nb}) \\ &= \frac{d}{dt}(I_g \vec{\omega}_{nb}) + \vec{\omega}_{nb} \times (I_g \vec{\omega}_{nb}) \\ &= I_g \dot{\vec{\omega}}_{nb} - (I_g \vec{\omega}_{nb}) \times \vec{\omega}_{nb}\end{aligned}$$

Rotational Motion about the CG Expressed in {b}

$$\mathbf{I}_g^b \dot{\boldsymbol{\omega}}_{nb} - \mathbf{S}(\mathbf{I}_g^b \boldsymbol{\omega}_{nb}) \boldsymbol{\omega}_{nb} = \mathbf{m}_g^b$$

where $\mathbf{I}_g^b$ is the *inertia dyadic*

where I_x , I_y , and I_z are the *moments of inertia* about {b} and $I_{xy}=I_{yx}$, $I_{xz}=I_{zx}$ and $I_{yz}=I_{zy}$ are the *products of inertia* defined as

$$\mathbf{I}_g^b := \begin{bmatrix} I_x & -I_{xy} & -I_{xz} \\ -I_{yx} & I_y & -I_{yz} \\ -I_{zx} & -I_{zy} & I_z \end{bmatrix}, \quad \mathbf{I}_g^b = (\mathbf{I}_g^b)^\top > 0$$

$$I_x = \int_V (y^2 + z^2) \rho_m dV;$$

$$I_{xy} = \int_V xy \rho_m dV = \int_V yx \rho_m dV = I_{yx}$$

$$I_y = \int_V (x^2 + z^2) \rho_m dV;$$

$$I_{xz} = \int_V xz \rho_m dV = \int_V zx \rho_m dV = I_{zx}$$

$$I_z = \int_V (x^2 + y^2) \rho_m dV;$$

$$I_{yz} = \int_V yz \rho_m dV = \int_V zy \rho_m dV = I_{zy}$$

3.1 质心的旋转运动

推导从欧拉的第二公理开始

$$\begin{aligned}
 \vec{m}_g &= \frac{d}{dt}(I_g \vec{\omega}_{ib}) \\
 &= \frac{d}{dt}(I_g \vec{\omega}_{nb}) \\
 &= \frac{d}{dt}(I_g \vec{\omega}_{nb}) + \vec{\omega}_{nb} \times (I_g \vec{\omega}_{nb}) \\
 &= I_g \dot{\vec{\omega}}_{nb} - (I_g \vec{\omega}_{nb}) \times \vec{\omega}_{nb}
 \end{aligned}$$

绕质心的旋转运动用 {b} 表示

$$I_g^b \dot{\vec{\omega}}_{nb} - S(I_g^b \vec{\omega}_{nb}) \vec{\omega}_{nb} = \vec{m}_g^b$$

inertia dyadic 在哪里

其中 I_x , I_y 和 I_z 是关于 {b} 的 *moments of inertia*, 而 $I_{xy}=I_{yx}$, $I_{xz}=I_{zx}$ 和 $I_{yz}=I_{zy}$ 是定义为 *products of inertia*

$$I_g^b := \begin{bmatrix} I_x & -I_{xy} & -I_{xz} \\ -I_{yx} & I_y & -I_{yz} \\ -I_{zx} & -I_{zy} & I_z \end{bmatrix}, \quad I_g^b = (I_g^b)^\top > 0$$

$$I_x = \int_V (y^2 + z^2) \rho_m dV;$$

$$I_{xy} = \int_V xy \rho_m dV = \int_V yx \rho_m dV = I_{yx}$$

$$I_y = \int_V (x^2 + z^2) \rho_m dV;$$

$$I_{xz} = \int_V xz \rho_m dV = \int_V zx \rho_m dV = I_{zx}$$

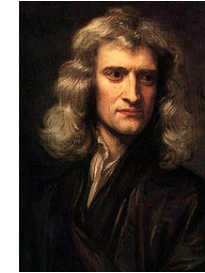
$$I_z = \int_V (x^2 + y^2) \rho_m dV;$$

$$I_{yz} = \int_V yz \rho_m dV = \int_V zy \rho_m dV = I_{zy}$$

3.1 Equations of Motion about the CG

The Newton-Euler equations can be represented in matrix form according to

$$\mathbf{M}_{RB}^{CG} \begin{bmatrix} \dot{\mathbf{v}}_{ng}^b \\ \dot{\boldsymbol{\omega}}_{nb}^b \end{bmatrix} + \mathbf{C}_{RB}^{CG} \begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \begin{bmatrix} \mathbf{f}_g^b \\ \mathbf{m}_g^b \end{bmatrix}$$



Isaac Newton (1642-1726)

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Expanding the matrices give

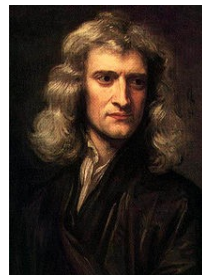
$$\underbrace{\begin{bmatrix} m\mathbf{I}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_g^b \end{bmatrix}}_{\mathbf{M}_{RB}^{CG}} \begin{bmatrix} \dot{\mathbf{v}}_{ng}^b \\ \dot{\boldsymbol{\omega}}_{nb}^b \end{bmatrix} + \underbrace{\begin{bmatrix} m\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{I}_g^b \boldsymbol{\omega}_{nb}^b) \end{bmatrix}}_{\mathbf{C}_{RB}^{CG}} \begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \begin{bmatrix} \mathbf{f}_g^b \\ \mathbf{m}_g^b \end{bmatrix}$$

$$\mathbf{M}_{RB} \dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu}) \boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

3.1 关于质心的运动方程

牛顿-欧拉方程可以根据矩阵形式表示

$$\mathbf{M}_{RB}^{CG} \begin{bmatrix} \dot{\mathbf{v}}_{ng}^b \\ \dot{\boldsymbol{\omega}}_{nb}^b \end{bmatrix} + \mathbf{C}_{RB}^{CG} \begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \begin{bmatrix} \mathbf{f}_g^b \\ \mathbf{m}_g^b \end{bmatrix}$$



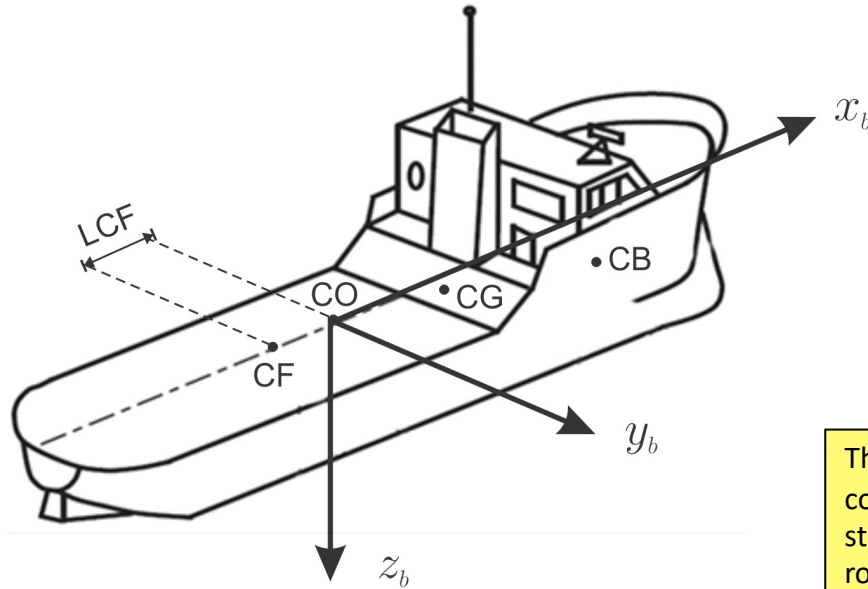
艾萨克·牛顿 (1642-1726) 莱昂哈德·欧拉 (1707-1783) [维基共享资源](#) [维基共享资源](#)

展开矩阵得到

$$\underbrace{\begin{bmatrix} m\mathbf{I}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_g^b \end{bmatrix}}_{\mathbf{M}_{RB}^{CG}} \begin{bmatrix} \dot{\mathbf{v}}_{ng}^b \\ \dot{\boldsymbol{\omega}}_{nb}^b \end{bmatrix} + \underbrace{\begin{bmatrix} m\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & -\mathbf{S}(\mathbf{I}_g^b \boldsymbol{\omega}_{nb}^b) \end{bmatrix}}_{\mathbf{C}_{RB}^{CG}} \begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \begin{bmatrix} \mathbf{f}_g^b \\ \mathbf{m}_g^b \end{bmatrix}$$

$$\mathbf{M}_{RB} \dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu}) \boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

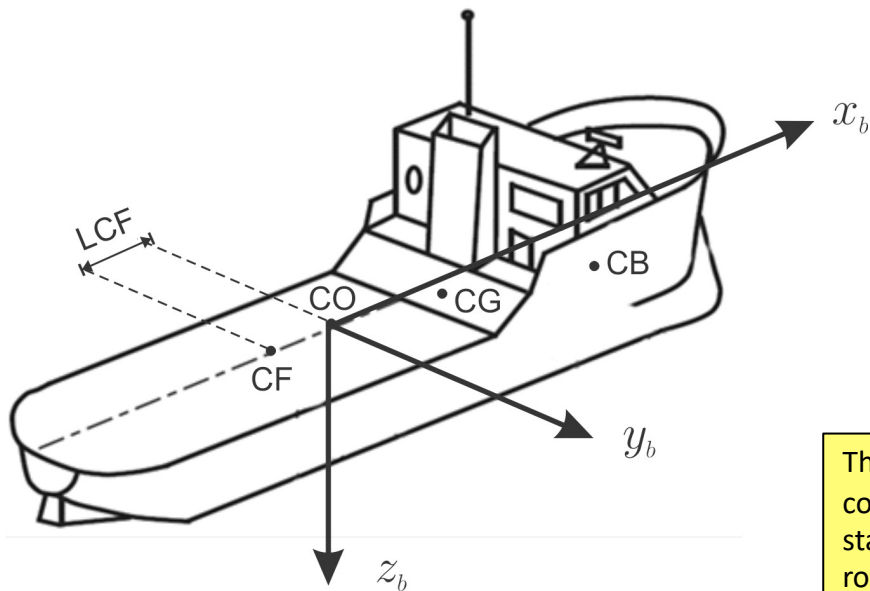
Which Coordinate Origin Should I Use?



It is recommended to represent the equations of motion in the origin of an arbitrary body-fixed Coordinate Origin (**CO**) instead of the Centre of Gravity (**CG**), which can be time-varying due to varying payload and fuel consumption.

The **CO** of the GNC system is chosen to satisfy the control objective. For instance, the **CO** can be either stabilized, acting as the point about which the craft rotates during stationkeeping, or directed to track a time-varying path in 2-D or 3-D space, serving as the point that follows the designated trajectory.

我应该使用哪个坐标原点？



建议将运动方程表示在任意刚体固定坐标原点（CO）处，而不是重心（CG），因为重心可能由于载荷变化和燃料消耗而随时间变化。

TheGNC 系统的 CO 选择以满足 control 目标。例如，CO 可以是 sta 被稳定，作为工艺的转折点 rot 在保持位置期间，或被指示在二维或三维空间中跟踪随时间变化的路径，作为跟随指定轨迹的点。

3.2 Newton-Euler Equations of Motion about the CO

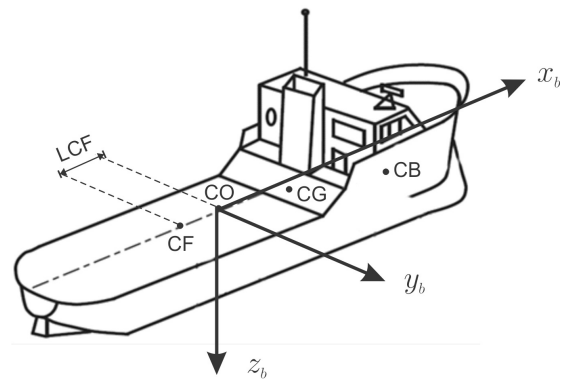
For marine craft, it is desirable to derive the equations of motion for an arbitrary Coordinate Origin (CO) usually chosen as coordinate origin of the GNC system.

The equations of motion can be transformed from the CG to the CO using the following coordinate transformation (see Appendix C)

$$\begin{aligned}
 \mathbf{v}_{ng}^b &= \mathbf{v}_{nb}^b + \boldsymbol{\omega}_{nb}^b \times \mathbf{r}_{bg}^b \\
 &= \mathbf{v}_{nb}^b - \mathbf{r}_{bg}^b \times \boldsymbol{\omega}_{nb}^b \\
 &= \mathbf{v}_{nb}^b + \mathbf{S}^\top(\mathbf{r}_{bg}^b) \boldsymbol{\omega}_{nb}^b
 \end{aligned}
 \Rightarrow
 \boxed{
 \begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{v}_{nb}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix}
 }$$

The transformation matrix is

$$\mathbf{H}(\mathbf{r}_{bg}^b) := \begin{bmatrix} \mathbf{I}_3 & \mathbf{S}^\top(\mathbf{r}_{bg}^b) \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_3 \end{bmatrix}, \quad \mathbf{H}^\top(\mathbf{r}_{bg}^b) = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_3 \end{bmatrix}$$



3.2 关于质心的牛顿-欧拉运动方程

对于海上船舶，最好为任意坐标原点（CO）推导运动方程，这通常选择为GNC系统的坐标原点。

可以使用以下坐标变换（见附录C）将运动方程从重心（CG）转换到质心（CO）

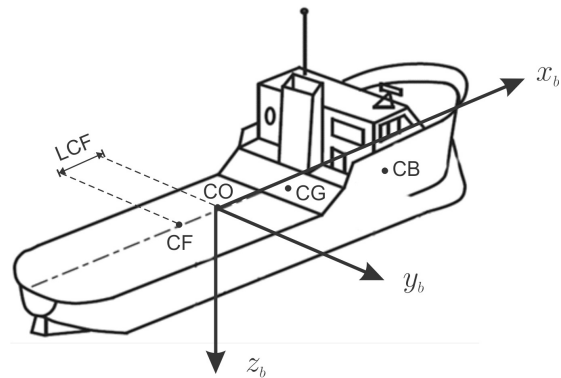
$$\begin{aligned}\mathbf{v}_{ng}^b &= \mathbf{v}_{nb}^b + \boldsymbol{\omega}_{nb}^b \times \mathbf{r}_{bg}^b \\ &= \mathbf{v}_{nb}^b - \mathbf{r}_{bg}^b \times \boldsymbol{\omega}_{nb}^b \\ &= \mathbf{v}_{nb}^b + \mathbf{S}^\top(\mathbf{r}_{bg}^b) \boldsymbol{\omega}_{nb}^b\end{aligned}$$



$$\begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{v}_{nb}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix}$$

T变换矩阵是

$$\mathbf{H}(\mathbf{r}_{bg}^b) := \begin{bmatrix} \mathbf{I}_3 & \mathbf{S}^\top(\mathbf{r}_{bg}^b) \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_3 \end{bmatrix}, \quad \mathbf{H}^\top(\mathbf{r}_{bg}^b) = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_3 \end{bmatrix}$$



3.2 Newton-Euler Equations of Motion about the CO

Newton-Euler equations in matrix-vector form about the CG (see Section 3.1)

$$\mathbf{M}_{RB}^{CG} \begin{bmatrix} \dot{\mathbf{v}}_{ng}^b \\ \dot{\boldsymbol{\omega}}_{nb}^b \end{bmatrix} + \mathbf{C}_{RB}^{CG} \begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \begin{bmatrix} \mathbf{f}_g^b \\ \mathbf{m}_g^b \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{v}_{nb}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix}$$

See App. C for more details

Newton-Euler equations in matrix-vector form about the CO

$$\mathbf{H}^\top(\mathbf{r}_{bg}^b) \mathbf{M}_{RB}^{CG} \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \dot{\mathbf{v}}_{nb}^b \\ \dot{\boldsymbol{\omega}}_{nb}^b \end{bmatrix} + \mathbf{H}^\top(\mathbf{r}_{bg}^b) \mathbf{C}_{RB}^{CG} \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{v}_{nb}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \mathbf{H}^\top(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{f}_g^b \\ \mathbf{m}_g^b \end{bmatrix}$$

$$\mathbf{M}_{RB} := \mathbf{H}^\top(\mathbf{r}_{bg}^b) \mathbf{M}_{RB}^{CG} \mathbf{H}(\mathbf{r}_{bg}^b)$$

$$\mathbf{C}_{RB} := \mathbf{H}^\top(\mathbf{r}_{bg}^b) \mathbf{C}_{RB}^{CG} \mathbf{H}(\mathbf{r}_{bg}^b)$$

Expanding the matrices

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_g^b - m\mathbf{S}^2(\mathbf{r}_{bg}^b) \end{bmatrix}$$

$$\mathbf{C}_{RB} = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & -m\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & -m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) - \mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b) \end{bmatrix}$$

It is possible to rewrite $\mathbf{M}_{RB}^{\{22\}}$ and $\mathbf{C}_{RB}^{\{22\}}$ using the Parallel-Axis Theorem. The motivation for this is to replace $\mathbf{I}_g$ with $\mathbf{I}_b$ (that is the inertia tensor in the CO instead of the CG).

3.2 关于质心的牛顿-欧拉运动方程

关于质心的牛顿-欧拉方程矩阵-向量形式（见第3.1节）

$$\mathbf{M}_{RB}^{CG} \begin{bmatrix} \dot{\mathbf{v}}_{ng}^b \\ \dot{\boldsymbol{\omega}}_{nb}^b \end{bmatrix} + \mathbf{C}_{RB}^{CG} \begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \begin{bmatrix} \mathbf{f}_g^b \\ \mathbf{m}_g^b \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{v}_{nb}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix}$$

详情请参见附录C

关于质心的牛顿-欧拉方程的矩阵-向量形式

$$\mathbf{H}^\top(\mathbf{r}_{bg}^b) \mathbf{M}_{RB}^{CG} \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \dot{\mathbf{v}}_{nb}^b \\ \dot{\boldsymbol{\omega}}_{nb}^b \end{bmatrix} + \mathbf{H}^\top(\mathbf{r}_{bg}^b) \mathbf{C}_{RB}^{CG} \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{v}_{nb}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \mathbf{H}^\top(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{f}_g^b \\ \mathbf{m}_g^b \end{bmatrix}$$

$$\mathbf{M}_{RB} := \mathbf{H}^\top(\mathbf{r}_{bg}^b) \mathbf{M}_{RB}^{CG} \mathbf{H}(\mathbf{r}_{bg}^b)$$

$$\mathbf{C}_{RB} := \mathbf{H}^\top(\mathbf{r}_{bg}^b) \mathbf{C}_{RB}^{CG} \mathbf{H}(\mathbf{r}_{bg}^b)$$

展开矩阵

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_g^b - m\mathbf{S}^2(\mathbf{r}_{bg}^b) \end{bmatrix}$$

$$\mathbf{C}_{RB} = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & -m\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & -m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) - \mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b) \end{bmatrix}$$

可以使用平行轴定理重写 $\mathbf{I}_{RB}^{\{22\}}$ 和 $\mathbf{I}_{RB}^{\{22\}}$ 这样做的动机是用 $\mathbf{I}_g$ （即质心处的惯性张量）替换 $\mathbf{I}_g$ 而不是重心处的 CG)。

3.2 Newton-Euler Equations of Motion about the CO

Theorem 3.1 (Huygens–Steiner's Parallel-Axis Theorem)

The inertia dyadic $\mathbf{I}_b^b = (\mathbf{I}_b^b)^\top \in \mathbb{R}^{3 \times 3}$ about the origin CO is

$$\begin{aligned}\mathbf{I}_b^b &= \mathbf{I}_g^b - m\mathbf{S}^2(\mathbf{r}_{bg}^b) \\ &= \mathbf{I}_g^b + m((\mathbf{r}_{bg}^b)^\top \mathbf{r}_{bg}^b \mathbf{I}_3 - \mathbf{r}_{bg}^b (\mathbf{r}_{bg}^b)^\top)\end{aligned}\quad (3.36)$$

where $\mathbf{I}_g^b = (\mathbf{I}_g^b)^\top \in \mathbb{R}^{3 \times 3}$ is the inertia dyadic about the CG. Expanding (3.36) gives

$$\mathbf{I}_b^b = \mathbf{I}_g^b + m \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -yx & x^2 + z^2 & -yz \\ -zx & -zy & x^2 + y^2 \end{bmatrix} \quad (3.37)$$

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b) & \underbrace{\mathbf{I}_g^b - m\mathbf{S}^2(\mathbf{r}_{bg}^b)}_{\mathbf{I}_b^b} \end{bmatrix}$$



Christian Huygens (1629-1695)

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3.2 关于质心的牛顿-欧拉运动方程

Theorem 3.1 (Huygens–Steiner's Parallel-Axis Theorem)

The inertia dyadic $\mathbf{I}_b^b = (\mathbf{I}_b^b)^\top \in \mathbb{R}^{3 \times 3}$ about the origin CO is

$$\begin{aligned}\mathbf{I}_b^b &= \mathbf{I}_g^b - m\mathbf{S}^2(\mathbf{r}_{bg}^b) \\ &= \mathbf{I}_g^b + m((\mathbf{r}_{bg}^b)^\top \mathbf{r}_{bg}^b \mathbf{I}_3 - \mathbf{r}_{bg}^b (\mathbf{r}_{bg}^b)^\top)\end{aligned}\quad (3.36)$$

where $\mathbf{I}_g^b = (\mathbf{I}_g^b)^\top \in \mathbb{R}^{3 \times 3}$ is the inertia dyadic about the CG. Expanding (3.36) gives

$$\mathbf{I}_b^b = \mathbf{I}_g^b + m \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -yx & x^2 + z^2 & -yz \\ -zx & -zy & x^2 + y^2 \end{bmatrix} \quad (3.37)$$

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b) & \underbrace{\mathbf{I}_g^b - m\mathbf{S}^2(\mathbf{r}_{bg}^b)}_{\mathbf{I}_b^b} \end{bmatrix}$$



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3.2 Newton-Euler Equations of Motion about the CO

$$C_{RB} = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & -m\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & -m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) - \mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b) \end{bmatrix}$$

Jacobi identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$$

Choosing $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ gives

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times (\mathbf{a} \times \mathbf{b})) &= -\mathbf{b} \times ((\mathbf{a} \times \mathbf{b}) \times \mathbf{a}) \\ &= ((\mathbf{a} \times \mathbf{b}) \times \mathbf{a}) \times \mathbf{b} \\ &= -(\mathbf{a} \times (\mathbf{a} \times \mathbf{b})) \times \mathbf{b} \end{aligned}$$

which is equivalent to

$$\mathbf{S}(\mathbf{a})\mathbf{S}(\mathbf{b})\mathbf{S}(\mathbf{a})\mathbf{b} = -\mathbf{S}(\mathbf{S}^2(\mathbf{a})\mathbf{b})\mathbf{b}$$

Let $\mathbf{a} = \mathbf{r}_{bg}^b$ and $\mathbf{b} = \boldsymbol{\omega}_{nb}^b$ such that

$$-\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b)\boldsymbol{\omega}_{nb}^b = \mathbf{S}(\mathbf{S}^2(\mathbf{r}_{bg}^b)\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b$$

The $C_{RB}^{\{22\}}$ element in (3.32) multiplied with $\boldsymbol{\omega}_{nb}^b$ is equivalent to

$$\begin{aligned} -[m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) + \mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b)]\boldsymbol{\omega}_{nb}^b \\ = m\mathbf{S}(\mathbf{S}^2(\mathbf{r}_{bg}^b)\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b - \mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b \\ = m\mathbf{S}(\mathbf{S}^2(\mathbf{r}_{bg}^b)\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b - \mathbf{S}([\mathbf{I}_g^b + m\mathbf{S}^2(\mathbf{r}_{bg}^b)]\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b \\ = -\mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b \end{aligned}$$

$$\mathbf{S}^\top(\mathbf{a})\mathbf{b} = -\mathbf{S}(\mathbf{a})\mathbf{b} = \mathbf{S}(\mathbf{b})\mathbf{a}.$$

3.2 关于质心的牛顿-欧拉运动方程

$$C_{RB} = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & -m\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b) & -m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) - \mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b) \end{bmatrix}$$

Jacobi identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$$

Choosing $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ gives

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times (\mathbf{a} \times \mathbf{b})) &= -\mathbf{b} \times ((\mathbf{a} \times \mathbf{b}) \times \mathbf{a}) \\ &= ((\mathbf{a} \times \mathbf{b}) \times \mathbf{a}) \times \mathbf{b} \\ &= -(\mathbf{a} \times (\mathbf{a} \times \mathbf{b})) \times \mathbf{b} \end{aligned}$$

which is equivalent to

$$\mathbf{S}(\mathbf{a})\mathbf{S}(\mathbf{b})\mathbf{S}(\mathbf{a})\mathbf{b} = -\mathbf{S}(\mathbf{S}^2(\mathbf{a})\mathbf{b})\mathbf{b}$$

Let $\mathbf{a} = \mathbf{r}_{bg}^b$ and $\mathbf{b} = \boldsymbol{\omega}_{nb}^b$ such that

$$-\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b)\boldsymbol{\omega}_{nb}^b = \mathbf{S}(\mathbf{S}^2(\mathbf{r}_{bg}^b)\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b$$

The $C_{RB}^{\{22\}}$ element in (3.32) multiplied with $\boldsymbol{\omega}_{nb}^b$ is equivalent to

$$\begin{aligned} -[m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{S}(\mathbf{r}_{bg}^b) + \mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b)]\boldsymbol{\omega}_{nb}^b \\ = m\mathbf{S}(\mathbf{S}^2(\mathbf{r}_{bg}^b)\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b - \mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b \\ = m\mathbf{S}(\mathbf{S}^2(\mathbf{r}_{bg}^b)\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b - \mathbf{S}([\mathbf{I}_g^b + m\mathbf{S}^2(\mathbf{r}_{bg}^b)]\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b \\ = -\mathbf{S}(\mathbf{I}_g^b\boldsymbol{\omega}_{nb}^b)\boldsymbol{\omega}_{nb}^b \end{aligned}$$

$$\mathbf{S}^\top(\mathbf{a})\mathbf{b} = -\mathbf{S}(\mathbf{a})\mathbf{b} = \mathbf{S}(\mathbf{b})\mathbf{a}.$$

3.2 Translational Motion about the CO

Translational Motion about the CO Expressed in $\{b\}$

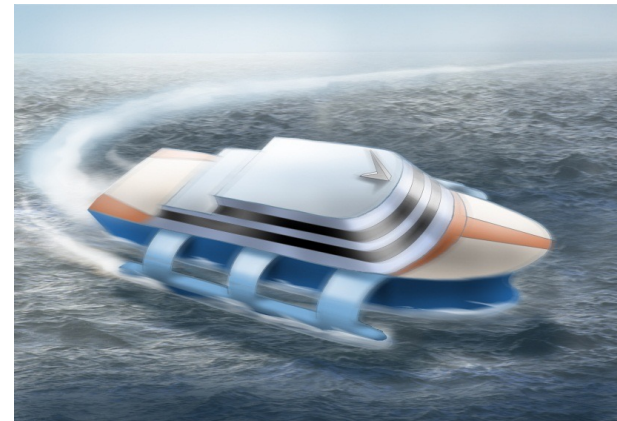
$$m[\dot{\mathbf{v}}_{nb}^b + \mathbf{S}(\dot{\boldsymbol{\omega}}_{nb}^b)\mathbf{r}_{bg}^b + \mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{v}_{nb}^b + \mathbf{S}^2(\boldsymbol{\omega}_{nb}^b)\mathbf{r}_{bg}^b] = \mathbf{f}_b^b$$

An alternative representation using vector cross products is

$$m[\dot{\mathbf{v}}_{nb}^b + \dot{\boldsymbol{\omega}}_{nb}^b \times \mathbf{r}_{bg}^b + \boldsymbol{\omega}_{nb}^b \times \mathbf{v}_{nb}^b + \boldsymbol{\omega}_{nb}^b \times (\boldsymbol{\omega}_{nb}^b \times \mathbf{r}_{bg}^b)] = \mathbf{f}_b^b$$



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3.2 关于质心的平动

关于 CO 的平动表达在 {b} 中

$$m[\dot{\mathbf{v}}_{nb}^b + \mathbf{S}(\dot{\boldsymbol{\omega}}_{nb}^b)\mathbf{r}_{bg}^b + \mathbf{S}(\boldsymbol{\omega}_{nb}^b)\mathbf{v}_{nb}^b + \mathbf{S}^2(\boldsymbol{\omega}_{nb}^b)\mathbf{r}_{bg}^b] = \mathbf{f}_b^b$$

使用向量叉乘的另一种表示是

$$m[\dot{\mathbf{v}}_{nb}^b + \dot{\boldsymbol{\omega}}_{nb}^b \times \mathbf{r}_{bg}^b + \boldsymbol{\omega}_{nb}^b \times \mathbf{v}_{nb}^b + \boldsymbol{\omega}_{nb}^b \times (\boldsymbol{\omega}_{nb}^b \times \mathbf{r}_{bg}^b)] = \mathbf{f}_b^b$$



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3.2 Rotational Motion about the CO

Rotational Motion about the CO Expressed in {b}

$$\mathbf{I}_b^b \dot{\boldsymbol{\omega}}_{nb}^b + \mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{I}_b^b \boldsymbol{\omega}_{nb}^b + m \mathbf{S}(\mathbf{r}_{bg}^b) \dot{\mathbf{v}}_{nb}^b + m \mathbf{S}(\mathbf{r}_{bg}^b) \mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{v}_{nb}^b = m \mathbf{b}^b$$

An alternative representation using vector cross products is

$$\mathbf{I}_b^b \dot{\boldsymbol{\omega}}_{nb}^b + \boldsymbol{\omega}_{nb}^b \times \mathbf{I}_b^b \boldsymbol{\omega}_{nb}^b + m \mathbf{r}_{bg}^b \times (\dot{\mathbf{v}}_{nb}^b + \boldsymbol{\omega}_{nb}^b \times \mathbf{v}_{nb}^b) = m \mathbf{b}^b$$

Theorem 3.1 (Huygens–Steiner’s Parallel-Axis Theorem)

The inertia dyadic $\mathbf{I}_b^b = (\mathbf{I}_b^b)^\top \in \mathbb{R}^{3 \times 3}$ about the origin CO is

$$\begin{aligned} \mathbf{I}_b^b &= \mathbf{I}_g^b - m \mathbf{S}^2(\mathbf{r}_{bg}^b) \\ &= \mathbf{I}_g^b + m ((\mathbf{r}_{bg}^b)^\top \mathbf{r}_{bg}^b \mathbf{I}_3 - \mathbf{r}_{bg}^b (\mathbf{r}_{bg}^b)^\top) \end{aligned} \quad (3.36)$$

where $\mathbf{I}_g^b = (\mathbf{I}_g^b)^\top \in \mathbb{R}^{3 \times 3}$ is the inertia dyadic about the CG. Expanding (3.36) gives

$$\mathbf{I}_b^b = \mathbf{I}_g^b + m \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -yx & x^2 + z^2 & -yz \\ -zx & -zy & x^2 + y^2 \end{bmatrix} \quad (3.37)$$



Christian Huygens (1629-1695)

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3.2 关于质心的旋转运动

绕 CO 的旋转运动用 $\{b\}$ 表示

$$\mathbf{I}_b^b \dot{\boldsymbol{\omega}}_{nb}^b + \mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{I}_b^b \boldsymbol{\omega}_{nb}^b + m \mathbf{S}(\mathbf{r}_{bg}^b) \dot{\mathbf{v}}_{nb}^b + m \mathbf{S}(\mathbf{r}_{bg}^b) \mathbf{S}(\boldsymbol{\omega}_{nb}^b) \mathbf{v}_{nb}^b = \mathbf{m}_b^b$$

使用向量叉乘的另一种表示是

$$\mathbf{I}_b^b \dot{\boldsymbol{\omega}}_{nb}^b + \boldsymbol{\omega}_{nb}^b \times \mathbf{I}_b^b \boldsymbol{\omega}_{nb}^b + m \mathbf{r}_{bg}^b \times (\dot{\mathbf{v}}_{nb}^b + \boldsymbol{\omega}_{nb}^b \times \mathbf{v}_{nb}^b) = \mathbf{m}_b^b$$

Theorem 3.1 (Huygens–Steiner’s Parallel-Axis Theorem)

The inertia dyadic $\mathbf{I}_b^b = (\mathbf{I}_b^b)^\top \in \mathbb{R}^{3 \times 3}$ about the origin CO is

$$\begin{aligned} \mathbf{I}_b^b &= \mathbf{I}_g^b - m \mathbf{S}^2(\mathbf{r}_{bg}^b) \\ &= \mathbf{I}_g^b + m ((\mathbf{r}_{bg}^b)^\top \mathbf{r}_{bg}^b \mathbf{I}_3 - \mathbf{r}_{bg}^b (\mathbf{r}_{bg}^b)^\top) \end{aligned} \quad (3.36)$$

where $\mathbf{I}_g^b = (\mathbf{I}_g^b)^\top \in \mathbb{R}^{3 \times 3}$ is the inertia dyadic about the CG. Expanding (3.36) gives

$$\mathbf{I}_b^b = \mathbf{I}_g^b + m \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -yx & x^2 + z^2 & -yz \\ -zx & -zy & x^2 + y^2 \end{bmatrix} \quad (3.37)$$



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3.3 Rigid-Body Equations of Motion

$\mathbf{f}_b^b = [X, Y, Z]^\top$	force through the CO expressed in $\{b\}$
$\mathbf{m}_b^b = [K, M, N]^\top$	moment about the CO expressed in $\{b\}$
$\mathbf{v}_{nb}^b = [u, v, w]^\top$	linear velocity of the CO relative to o_n expressed in $\{b\}$
$\boldsymbol{\omega}_{nb}^b = [p, q, r]^\top$	angular velocity of $\{b\}$ relative to $\{n\}$ expressed in $\{b\}$
$\mathbf{r}_{bg}^b = [x_g, y_g, z_g]^\top$	vector from the CO to the CG expressed in $\{b\}$

Component Form (SNAME 1950)

$$\begin{aligned}
 m [\dot{u} - vr + wq - x_g(q^2 + r^2) + y_g(pq - \dot{r}) + z_g(pr + \dot{q})] &= X \\
 m [\dot{v} - wp + ur - y_g(r^2 + p^2) + z_g(qr - \dot{p}) + x_g(qp + \dot{r})] &= Y \\
 m [\dot{w} - uq + vp - z_g(p^2 + q^2) + x_g(rp - \dot{q}) + y_g(rq + \dot{p})] &= Z \\
 I_x \dot{p} + (I_z - I_y)qr - (\dot{r} + pq)I_{xz} + (r^2 - q^2)I_{yz} + (pr - \dot{q})I_{xy} \\
 + m [y_g(\dot{w} - uq + vp) - z_g(\dot{v} - wp + ur)] &= K \\
 I_y \dot{q} + (I_x - I_z)rp - (\dot{p} + qr)I_{xy} + (p^2 - r^2)I_{zx} + (qp - \dot{r})I_{yz} \\
 + m [z_g(\dot{u} - vr + wq) - x_g(\dot{w} - uq + vp)] &= M \\
 I_z \dot{r} + (I_y - I_x)pq - (\dot{q} + rp)I_{yz} + (q^2 - p^2)I_{xy} + (rq - \dot{p})I_{zx} \\
 + m [x_g(\dot{v} - wp + ur) - y_g(\dot{u} - vr + wq)] &= N
 \end{aligned}$$

These are the 6-DOF rigid-body equations of motion commonly found in textbooks.

The upcoming formulas will provide an alternative *matrix-vector* representation.

3.3 刚体运动方程

$\mathbf{f}_b^b = [X, Y, Z]^\top$	force through the CO expressed in $\{b\}$
$\mathbf{m}_b^b = [K, M, N]^\top$	moment about the CO expressed in $\{b\}$
$\mathbf{v}_{nb}^b = [u, v, w]^\top$	linear velocity of the CO relative to o_n expressed in $\{b\}$
$\boldsymbol{\omega}_{nb}^b = [p, q, r]^\top$	angular velocity of $\{b\}$ relative to $\{n\}$ expressed in $\{b\}$
$\mathbf{r}_{bg}^b = [x_g, y_g, z_g]^\top$	vector from the CO to the CG expressed in $\{b\}$

组件形式 (SNAME 1950)

$$\begin{aligned}
 m [\dot{u} - vr + wq - x_g(q^2 + r^2) + y_g(pq - \dot{r}) + z_g(pr + \dot{q})] &= X \\
 m [\dot{v} - wp + ur - y_g(r^2 + p^2) + z_g(qr - \dot{p}) + x_g(qp + \dot{r})] &= Y \\
 m [\dot{w} - uq + vp - z_g(p^2 + q^2) + x_g(rp - \dot{q}) + y_g(rq + \dot{p})] &= Z \\
 I_x \dot{p} + (I_z - I_y)qr - (\dot{r} + pq)I_{xz} + (r^2 - q^2)I_{yz} + (pr - \dot{q})I_{xy} \\
 + m [y_g(\dot{w} - uq + vp) - z_g(\dot{v} - wp + ur)] &= K \\
 I_y \dot{q} + (I_x - I_z)rp - (\dot{p} + qr)I_{xy} + (p^2 - r^2)I_{zx} + (qp - \dot{r})I_{yz} \\
 + m [z_g(\dot{u} - vr + wq) - x_g(\dot{w} - uq + vp)] &= M \\
 I_z \dot{r} + (I_y - I_x)pq - (\dot{q} + rp)I_{yz} + (q^2 - p^2)I_{xy} + (rq - \dot{p})I_{zx} \\
 + m [x_g(\dot{v} - wp + ur) - y_g(\dot{u} - vr + wq)] &= N
 \end{aligned}$$

这些是教材中常见的六自由度刚体运动方程。

即将推出的公式将提供一种替代的 *matrix-vector* 表示方式。

3.3 Rigid-Body Equations of Motion

Matrix-Vector Form

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

$$\begin{aligned}\mathbf{v} &= [u, v, w, p, q, r]^\top && \text{Generalized velocity} \\ \boldsymbol{\tau}_{RB} &= [X, Y, Z, K, M, N]^\top && \text{Generalized force}\end{aligned}$$

Property 3.1 (Rigid-Body System Inertia Matrix)

$$\begin{aligned}\mathbf{M}_{RB} &= \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_b^b \end{bmatrix} \\ &= \begin{bmatrix} m & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m & 0 & -mz_g & 0 & mx_g \\ 0 & 0 & m & my_g & -mx_g & 0 \\ 0 & -mz_g & my_g & I_x & -I_{xy} & -I_{xz} \\ mz_g & 0 & -mx_g & -I_{yx} & I_y & -I_{yz} \\ -my_g & mx_g & 0 & -I_{zx} & -I_{zy} & I_z \end{bmatrix}\end{aligned}$$

$$\mathbf{M}_{RB} = \mathbf{M}_{RB}^\top > 0, \quad \dot{\mathbf{M}}_{RB} = \mathbf{0}_{6 \times 6}$$

3.3 刚体运动方程

矩阵-向量形式

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

!! !u! v! w! p! q! r “ τ 广义速度 广
义力 $\mathbf{X}, \dot{\mathbf{Y}}, \mathbf{Z}, \mathbf{K}, \mathbf{M}, \mathbf{N}]^T$

属性 3.1 (刚体系统惯性矩阵)

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_b^b \end{bmatrix}$$

$$= \begin{bmatrix} m & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m & 0 & -mz_g & 0 & mx_g \\ 0 & 0 & m & my_g & -mx_g & 0 \\ 0 & -mz_g & my_g & I_x & -I_{xy} & -I_{xz} \\ mz_g & 0 & -mx_g & -I_{yx} & I_y & -I_{yz} \\ -my_g & mx_g & 0 & -I_{zx} & -I_{zy} & I_z \end{bmatrix}$$

$$\mathbf{M}_{RB} = \mathbf{M}_{RB}^T > 0, \quad \dot{\mathbf{M}}_{RB} = \mathbf{0}_{6 \times 6}$$

3.3 Rigid-Body Equations of Motion

Matlab:

The rigid-body system inertia matrix M_{RB} can be computed in Matlab as

```
r_g = [10 0 1]';    % location of the CG with respect to the CO
R44 = 10;            % radius of gyration in roll
R55 = 20;            % radius of gyration in pitch
R66 = 5;             % radius of gyration in yaw
m = 1000;            % mass
I_g = m * diag([R44^2 R55^2 R66^2]);    % inertia dyadic (CG)

% rigid-body system inertia matrix
S = Smtrx(r_g);
MRB = [ m * eye(3)   -m * S
       m * S         I_g - m * S^2 ]

MRB =

    1000         0         0         0    1000         0
         0    1000         0   -1000         0   10000
         0         0    1000         0  -10000         0
         0   -1000         0  101000         0  -10000
    1000         0  -10000         0  501000         0
         0   10000         0  -10000         0  125000
```

The rigid-body system inertia matrix can also be computed using the command

```
MRB = rbody(m, R44, R55, R66, zeros(3,1), r_g)
```

3.3 刚体运动方程

Matlab:

The rigid-body system inertia matrix M_{RB} can be computed in Matlab as

```
r_g = [10 0 1]';    % location of the CG with respect to the CO
R44 = 10;            % radius of gyration in roll
R55 = 20;            % radius of gyration in pitch
R66 = 5;             % radius of gyration in yaw
m = 1000;            % mass
I_g = m * diag([R44^2 R55^2 R66^2]);    % inertia dyadic (CG)

% rigid-body system inertia matrix
S = Smtrx(r_g);
MRB = [ m * eye(3)   -m * S
       m * S         I_g - m * S^2 ]

MRB =

    1000         0         0         0    1000         0
         0    1000         0   -1000         0   10000
         0         0    1000         0  -10000         0
         0   -1000         0  101000         0  -10000
    1000         0  -10000         0  501000         0
         0   10000         0  -10000         0  125000
```

The rigid-body system inertia matrix can also be computed using the command

```
MRB = rbody(m, R44, R55, R66, zeros(3,1), r_g)
```

3.3 Rigid-Body Equations of Motion

Theorem 3.2 (Coriolis-Centripetal Matrix from System Inertia Matrix)

Let $\mathbf{M}$ be a 6×6 *system inertia matrix* defined as:

$$\mathbf{M} = \mathbf{M}^\top = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix} > 0$$

where $\mathbf{M}_{21} = \mathbf{M}_{12}^\top$. Then the *Coriolis-centripetal matrix* can always be parameterized such that

$$\mathbf{C}(\mathbf{v}) = -\mathbf{C}^\top(\mathbf{v})$$

$$\mathbf{C}(\mathbf{v}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & \mathbf{S}(\mathbf{M}_{11}\mathbf{v}_1 + \mathbf{M}_{12}\mathbf{v}_2) \\ \mathbf{S}(\mathbf{M}_{11}\mathbf{v}_1 + \mathbf{M}_{12}\mathbf{v}_2) & \mathbf{S}(\mathbf{M}_{21}\mathbf{v}_1 + \mathbf{M}_{22}\mathbf{v}_2) \end{bmatrix}$$

where $\mathbf{v}_1 = [u, v, w]^\top$, $\mathbf{v}_2 = [p, q, r]^\top$

Proof: Sagatun and Fossen (1991).

Sagatun, S. I. and T. I. Fossen (1991). Lagrangian Formulation of Underwater Vehicles' Dynamics. *IEEE International Conference on Systems, Man, and Cybernetics*, Charlottesville, VA, USA, IEEE Xplore, pp. 1029-1034.

3.3 刚体运动方程

定理 3.2 （由系统惯性矩阵得到的科里奥利-向心矩阵）

设 $\mathbf{M}$ 为一个 6×6 *system inertia matrix*, 定义如下:

$$\mathbf{M} = \mathbf{M}^T = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix} > 0$$

其中 $\mathbf{M}_{21} = \mathbf{M}_{12}^T$ 。然后 *Coriolis-centripetal matrix* 总是可以被参数化, 使得

$$\mathbf{C}(\mathbf{v}) = -\mathbf{C}^T(\mathbf{v})$$

$$\mathbf{C}(\mathbf{v}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & \mathbf{S}(\mathbf{M}_{11}\mathbf{v}_1 + \mathbf{M}_{12}\mathbf{v}_2) \\ \mathbf{S}(\mathbf{M}_{11}\mathbf{v}_1 + \mathbf{M}_{12}\mathbf{v}_2) & \mathbf{S}(\mathbf{M}_{21}\mathbf{v}_1 + \mathbf{M}_{22}\mathbf{v}_2) \end{bmatrix}$$

哪里 !!! !u “v” w “τ” ! 2 “! p” q “r” τ

证据: Sagatun 和 Fossen (1991)。

Sagatun, S. I. 和 T. I. Fossen (1991)。水下车辆动力学的拉格朗日公式。IEEE International Conference on Systems, Man, and Cybernetics, 弗吉尼亚州夏洛茨维尔, 美国, IEEE Xplore, 第1029-1034页。

3.3 Rigid-Body Equations of Motion

Property 3.2 (Rigid-Body Coriolis and Centripetal Matrix)

The *rigid-body Coriolis and centripetal matrix* $\mathbf{C}_{RB}(\mathbf{v})$ can always be represented such that $\mathbf{C}_{RB}(\mathbf{v})$ is skew-symmetric. This is mathematically equivalent to

$$\mathbf{C}_{RB}(\mathbf{v}) = -\mathbf{C}_{RB}^T(\mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{R}^6$$

The skew-symmetric property is very useful when designing nonlinear motion control system since the quadratic form

$$\mathbf{v}^T \mathbf{C}_{RB}(\mathbf{v}) \mathbf{v} \equiv 0$$

This is exploited in energy-based control designs where Lyapunov functions play a key role. The same property is also used in nonlinear observer design.

There exist several parameterizations that satisfy Property 3.2. Two important ones are the:

- Lagrangian parametrization
- Linear velocity-independent parameterization

which are presented on the forthcoming pages.

3.3 刚体运动方程

属性 3.2 (刚体科里奥利力和向心力矩阵)

rigid-body Coriolis and centripetal matrix 总是可以被表示为

那是斜对称的。这在数学上等价于

$$\mathbf{C}_{RB}(\mathbf{v}) = -\mathbf{C}_{RB}^T(\mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{R}^6$$

斜对称性质在设计非线性运动控制系统时非常有用，因为二次型 $\mathbf{v}^T \mathbf{C}_{RB}(\mathbf{v}) \mathbf{v} \equiv 0$

这在基于能量的控制设计中得到了利用，其中李雅普诺夫函数起着关键作用。相同的特性也用于非线性观测器设计。

存在几种满足性质 3.2 的参数化。其中两个重要的是：

- 拉格朗日参数化
- 线速度无关参数化

将会在接下来的页面中呈现的。

3.3 Rigid-Body Equations of Motion

Lagrangian Parameterization

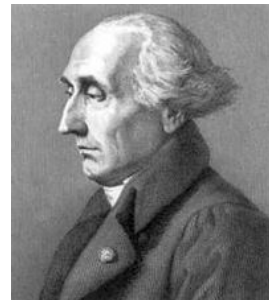
Application of the Theorem 3.2 with $\mathbf{M} = \mathbf{M}_{RB}$ yields the following expression

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_{bg}^b) \\ -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_{bg}^b) & m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_1)\mathbf{r}_{bg}^b) - \mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

which can be rewritten according to

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_{bg}^b) \\ -m\mathbf{S}(\boldsymbol{\nu}_1) + m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

To ensure that $\mathbf{C}_{RB}(\mathbf{v}) = -\mathbf{C}_{RB}(\mathbf{v})^T$, it is necessary to use $\mathbf{S}(\mathbf{v}_1)\mathbf{v}_1 = \mathbf{0}$ and add $\mathbf{S}(\mathbf{v}_1)$ in $\mathbf{C}_{RB}^{\{21\}}$



Joseph-Louis Lagrange (1736-1813)

[Wikimedia Commons](#)

Sagatun, S. I. and T. I. Fossen (1991). Lagrangian Formulation of Underwater Vehicles' Dynamics. *IEEE International Conference on Systems, Man, and Cybernetics*, Charlottesville, VA, USA, IEEE Xplore, pp. 1029-1034.

3.3 刚体运动方程

拉格朗日参数化

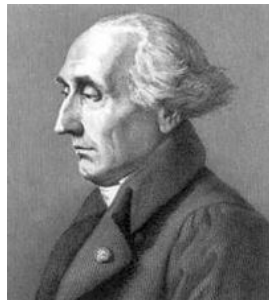
将定理 3.2 应用于 $\mathbf{M} = \mathbf{M}_{RB}$ 得到以下表达式

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_{bg}^b) \\ -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_{bg}^b) & m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_1)\mathbf{r}_{bg}^b) - \mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

可以根据……重写

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_{bg}^b) \\ -m\mathbf{S}(\boldsymbol{\nu}_1) + m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

为了确保 $\mathbf{C}_{RB}(\mathbf{v}) = -\mathbf{C}_{RB}(\mathbf{v})^T$, 有必要使用 $\mathbf{S}(\mathbf{v})\mathbf{v} = 0$ 并在 $\mathbf{C}_{RB}^{(21)}$ 中加入 $\mathbf{S}(\mathbf{v})$



约瑟夫-路易·拉格朗日 (1736-1813)
[\) 维基共享资源](#)

Sagatun, S. I. and T. I. Fossen (1991). Lagrangian Formulation of Underwater Vehicles' Dynamics. *IEEE International Conference on Systems, Man, and Cybernetics*, Charlottesville, VA, USA, IEEE Xplore, pp. 1029-1034.

讲义：海洋船舶流体动力学与运动控制 (T. I. Fossen)

3.3 Rigid-Body Equations of Motion

Lagrangian Parameterization

$$C_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_{bg}^b) \\ -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_{bg}^b) & m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_1)\mathbf{r}_{bg}^b) - \mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

Component Form

$$C_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -m(y_g q + z_g r) & m(y_g p + w) & m(z_g p - v) & m(y_g q + z_g r) & -m(x_g q - w) & -m(x_g r + v) \\ m(x_g q - w) & -m(z_g r + x_g p) & m(z_g q + u) & -m(y_g p + w) & m(z_g r + x_g p) & -m(y_g r - u) \\ m(x_g r + v) & m(y_g r - u) & -m(x_g p + y_g q) & -m(z_g p - v) & -m(z_g q + u) & m(x_g p + y_g q) \\ m(y_g q + z_g r) & -m(x_g q - w) & -m(x_g r + v) & 0 & -I_{yz}q - I_{xz}p + I_z r & I_{yz}r + I_{xy}p - I_y q \\ -m(y_g p + w) & m(z_g r + x_g p) & -m(y_g r - u) & I_{yz}q + I_{xz}p - I_z r & 0 & -I_{xz}r - I_{xy}q + I_x p \\ -m(z_g p - v) & -m(z_g q + u) & m(x_g p + y_g q) & -I_{yz}r - I_{xy}p + I_y q & I_{xz}r + I_{xy}q - I_x p & 0 \end{bmatrix} \quad (3.59)$$

3.3 刚体运动方程

拉格朗日参数化

$$C_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_{bg}^b) \\ -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_{bg}^b) & m\mathbf{S}(\mathbf{S}(\boldsymbol{\nu}_1)\mathbf{r}_{bg}^b) - \mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

组件形式

$$C_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -m(y_g q + z_g r) & m(y_g p + w) & m(z_g p - v) \\ m(x_g q - w) & -m(z_g r + x_g p) & m(z_g q + u) \\ m(x_g r + v) & m(y_g r - u) & -m(x_g p + y_g q) \\ m(y_g q + z_g r) & -m(x_g q - w) & -m(x_g r + v) \\ -m(y_g p + w) & m(z_g r + x_g p) & -m(y_g r - u) \\ -m(z_g p - v) & -m(z_g q + u) & m(x_g p + y_g q) \\ 0 & -I_{yz}q - I_{xz}p + I_z r & I_{yz}r + I_{xy}p - I_y q \\ I_{yz}q + I_{xz}p - I_z r & 0 & -I_{xz}r - I_{xy}q + I_x p \\ -I_{yz}r - I_{xy}p + I_y q & I_{xz}r + I_{xy}q - I_x p & 0 \end{bmatrix} \quad (3.59)$$

3.3 Rigid-Body Equations of Motion

Linear Velocity-Independent Parameterization

By using the cross-product property $\mathbf{S}(\mathbf{v}_1)\mathbf{v}_2 = -\mathbf{S}(\mathbf{v}_2)\mathbf{v}_1$, it is possible to move $\mathbf{S}(\mathbf{v}_1)\mathbf{v}_2$ from $\mathbf{C}_{RB}^{\{12\}}$ to $\mathbf{C}_{RB}^{\{11\}}$. This gives an expression for $\mathbf{C}_{RB}(\mathbf{v})$ that is independent of linear velocity $\mathbf{v}_1$ (Fossen and Fjellstad 1995)

$$\mathbf{C}_{RB}(\mathbf{v}) = \begin{bmatrix} m\mathbf{S}(\mathbf{v}_2) & -m\mathbf{S}(\mathbf{v}_2)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\mathbf{v}_2) & -\mathbf{S}(\mathbf{I}_b^b\mathbf{v}_2) \end{bmatrix}$$

This formula is the preferred representation when ocean currents enter the equations of motion. The main reason is that $\mathbf{C}_{RB}(\mathbf{v})$ does not depend on the linear velocity vector $\mathbf{v}_1 = [u, v, w]^T$. For a marine craft exposed to irrotational ocean currents, it follows from Property 10.1 in Section 10.3 that

$$\mathbf{M}_{RB}\dot{\mathbf{v}} + \mathbf{C}_{RB}(\mathbf{v})\mathbf{v} \equiv \mathbf{M}_{RB}\dot{\mathbf{v}}_r + \mathbf{C}_{RB}(\mathbf{v}_r)\mathbf{v}_r$$

where the relative velocity vector $\mathbf{v}_r = \mathbf{v} - \mathbf{v}_c$ is defined such that only linear ocean current velocities are used

$$\mathbf{v}_c = [u_c, v_c, w_c, 0, 0, 0]^T$$

Fossen, T. I. and O.-E. Fjellstad (1995). Nonlinear Modelling of Marine Vehicles in 6 Degrees of Freedom. *International Journal of Mathematical Modelling of Systems* 1(1), 17-27.

3.3 刚体运动方程

线速度无关参数化

通过使用交叉乘积性质 $S(\mathbf{v})\mathbf{v} = -S(\mathbf{v})\mathbf{v}$ ，可以将 $S(\mathbf{v})\mathbf{v}$ 从 $C_{RB}^{(12)}$ 移动到 $C_{RB}^{(11)}$ 。这给出了一个与线速度 $\mathbf{v}$ 无关的 $C_{RB}(\mathbf{v})$ 表达式 (Fossen 和 Fjellstad 1995)

$$C_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & -m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

当海流进入运动方程时，这个公式是首选表示方法。主要原因是 $C_{RB}(\mathbf{v})$ 不依赖于线速度向量 $\mathbf{v} = [u, v, w]^T$ 。对于暴露在无旋海流中的海上航行器，根据第10.3节中的性质10.1可知

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} \equiv \mathbf{M}_{RB}\dot{\boldsymbol{\nu}}_r + \mathbf{C}_{RB}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r$$

其中相对速度矢量 $\boldsymbol{\nu}_r = \mathbf{v} - \mathbf{v}_c$ 的定义仅使用线性洋流速度

$$\boldsymbol{\nu}_c = [u_c, v_c, w_c, 0, 0, 0]^T$$

Fossen, T. I. 和 O.-E. Fjellstad (1995)。海洋船舶六自由度非线性建模。《v22》 《v23》 1(1), 17-27。

3.3 Rigid-Body Equations of Motion

Linear Velocity-Independent Parameterization

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & -m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

This formula can also be expressed in terms of the $\mathbf{C}_{RB}$ matrix in the CG, and the transformation matrix from the CG to the CO

$$\begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{v}_{nb}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix}$$

See Appendix C for more details

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \mathbf{H}^\top(\mathbf{r}_{bg}^b) \begin{bmatrix} 0 & -mr & mq & 0 & 0 & 0 \\ mr & 0 & -mp & 0 & 0 & 0 \\ -mq & mp & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_z r & -I_y q \\ 0 & 0 & 0 & -I_z r & 0 & I_x p \\ 0 & 0 & 0 & I_y q & -I_x p & 0 \end{bmatrix} \mathbf{H}(\mathbf{r}_{bg}^b)$$

3.3 刚体运动方程

线速度无关参数化

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & -m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

这个公式也可以用CG中的 $\mathbf{C}_{RB}$ 矩阵以及从CG到CO的变换矩阵来表示

$$\begin{bmatrix} \mathbf{v}_{ng}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix} = \mathbf{H}(\mathbf{r}_{bg}^b) \begin{bmatrix} \mathbf{v}_{nb}^b \\ \boldsymbol{\omega}_{nb}^b \end{bmatrix}$$

更多详情请参见附录C

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \mathbf{H}^\top(\mathbf{r}_{bg}^b) \begin{bmatrix} 0 & -mr & mq & 0 & 0 & 0 \\ mr & 0 & -mp & 0 & 0 & 0 \\ -mq & mp & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_z r & -I_y q \\ 0 & 0 & 0 & -I_z r & 0 & I_x p \\ 0 & 0 & 0 & I_y q & -I_x p & 0 \end{bmatrix} \mathbf{H}(\mathbf{r}_{bg}^b)$$

3.3 Rigid-Body Equations of Motion

Matlab:

The Lagrangian parametrization (Theorem 3.2) is implemented in the Matlab MSS toolbox in the function `m2c.m`, while the linear-velocity independent parametrization (3.63) is implemented in the more generic function `rbody.m`. The following example demonstrates how $C_{RB}(\nu)$ can be computed numerically

```
r_g = [10 0 1]'; % location of the CG relative to the CO
R44 = 10; % radius of gyration in roll
R55 = 20; % radius of gyration in pitch
R66 = 5; % radius of gyration in yaw
m = 1000; % mass
nu = [8 0.5 0.1 0.2 -0.3 0.2]'; % velocity vector
```

% Method 1: Linear velocity-independent parametrization

```
nu2 = nu(4:6);
[MRB,CRB] = rbody(m, R44, R55, R66, nu2, r_g)
```

```
MRB =
    1000         0         0         0    1000         0
         0    1000         0   -1000         0   10000
         0         0    1000         0  -10000         0
         0   -1000         0  101000         0  -10000
    1000         0  -10000         0  501000         0
         0   10000         0  -10000         0  125000

CRB =
         0   -200   -300    200    3000   -2000
    200         0   -200         0    2200         0
    300    200         0   -200    300    2000
   -200         0    200         0    2800  120000
  -3000  -2200   -300   -2800         0   -2000
    2000         0  -2000  -120000    2000         0
```

% Method 2: Lagrangian parametrization

```
CRB = m2c(MRB, nu)
```

```
CRB =
         0         0         0         0    3100   -2300
         0         0         0   -3100         0    7700
         0         0         0    2300   -7700         0
         0    3100   -2300         0   28000  143300
   -3100         0    7700   -28000         0   17700
    2300   -7700         0  -143300  -17700         0
```

Even though the numerical values for the two $C_{RB}(\nu)$ matrices are different, they both produce the same product $C_{RB}(\nu)\nu$.

3.3 刚体运动方程

Matlab:

The Lagrangian parametrization (Theorem 3.2) is implemented in the Matlab MSS toolbox in the function `m2c.m`, while the linear-velocity independent parametrization (3.63) is implemented in the more generic function `rbody.m`. The following example demonstrates how $C_{RB}(\nu)$ can be computed numerically

```
r_g = [10 0 1]'; % location of the CG relative to the CO
R44 = 10; % radius of gyration in roll
R55 = 20; % radius of gyration in pitch
R66 = 5; % radius of gyration in yaw
m = 1000; % mass
nu = [8 0.5 0.1 0.2 -0.3 0.2]'; % velocity vector
```

```
% Method 1: Linear velocity-independent parametrization
nu2 = nu(4:6);
[MRB,CRB] = rbody(m, R44, R55, R66, nu2, r_g)
```

```
MRB =
    1000         0         0         0    1000         0
         0    1000         0   -1000         0   10000
         0         0    1000         0  -10000         0
         0   -1000         0  101000         0  -10000
    1000         0  -10000         0  501000         0
         0   10000         0  -10000         0  125000

CRB =
         0   -200   -300    200    3000   -2000
        200         0   -200         0    2200         0
        300    200         0   -200    300    2000
       -200         0    200         0    2800  120000
      -3000  -2200   -300   -2800         0   -2000
        2000         0  -2000  -120000    2000         0
```

```
% Method 2: Lagrangian parametrization
CRB = m2c(MRB, nu)
```

```
CRB =
         0         0         0         0    3100   -2300
         0         0         0   -3100         0    7700
         0         0         0    2300   -7700         0
         0    3100   -2300         0   28000  143300
       -3100         0    7700   -28000         0   17700
        2300   -7700         0  -143300  -17700         0
```

Even though the numerical values for the two $C_{RB}(\nu)$ matrices are different, they both produce the same product $C_{RB}(\nu)\nu$.

3.3 Linearized 6-DOF Rigid-Body Equations of Motion

The nonlinear rigid-body equations of motion

$$\mathbf{M}_{RB}\dot{\mathbf{v}} + \mathbf{C}_{RB}(\mathbf{v})\mathbf{v} = \boldsymbol{\tau}_{RB}$$

can be linearized about $\mathbf{v}_0 = [U, 0, 0, 0, 0, 0]^T$ for a marine craft moving at forward speed U .

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}^*\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

$$\mathbf{C}_{RB}^* = U\mathbf{M}_{RB}\mathbf{L}, \quad \mathbf{C}_{RB}^* \neq -(\mathbf{C}_{RB}^*)^T$$

$$\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The linearized Coriolis and centripetal forces are recognized as

$$\mathbf{f}_c = \mathbf{C}_{RB}^*\boldsymbol{\nu} = \begin{bmatrix} 0 \\ mUr \\ -mUq \\ -m y_g Uq - m z_g Ur \\ m x_g Uq \\ m x_g Ur \end{bmatrix}$$

3.3 线性化六自由度刚体运动方程

非线性刚体运动方程

$$\dot{\mathbf{v}}_{RB} = \mathbf{C}_{RB}^* \mathbf{v}_{RB} + \mathbf{g}_{RB}$$

可以关于 $\mathbf{v}_0 = [U, 0, 0, 0, 0, 0]^T$ 对沿前进速度 U 运动的海洋船舶进行线性化。

$$\mathbf{M}_{RB} \dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}^* \boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

$$\mathbf{C}_{RB}^* = U \mathbf{M}_{RB} \mathbf{L}, \quad \mathbf{C}_{RB}^* \neq -(\mathbf{C}_{RB}^*)^T$$

线性化的科里奥利力和向心力被认为是

$$\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{f}_c = \mathbf{C}_{RB}^* \boldsymbol{\nu} = \begin{bmatrix} 0 \\ mUr \\ -mUq \\ -m y_g Uq - m z_g Ur \\ m x_g Uq \\ m x_g Ur \end{bmatrix}$$

3.3 Linearized 6-DOF Rigid-Body Equations of Motion

Matlab:

The linearized model (3.67) is computed using the following Matlab commands

```
U = 1;
MRB = [ 1000      0      0      0      1000      0
        0      1000      0     -1000      0     10000
        0      0      1000      0     -10000      0
        0     -1000      0     101000      0     -10000
       1000      0     -10000      0     501000      0
        0     10000      0     -10000      0     125000];

L = zeros(6,6); L(2,6) = 1; L(3,5) = -1;
CRB = U * MRB * L

CRB =
      0      0      0      0      0      0
      0      0      0      0      0     1000
      0      0      0      0     -1000      0
      0      0      0      0      0     -1000
      0      0      0      0     10000      0
      0      0      0      0      0     10000
```

Note that the skew-symmetric property is destroyed by linearization. In other words, $\mathbf{C}_{\text{RB}}^* \neq -(\mathbf{C}_{\text{RB}}^*)^\top$

3.3 线性化六自由度刚体运动方程

Matlab:

The linearized model (3.67) is computed using the following Matlab commands

```
U = 1;
MRB = [ 1000      0      0      0      1000      0
         0      1000      0     -1000      0     10000
         0      0      1000      0     -10000      0
         0     -1000      0    101000      0     -10000
        1000      0     -10000      0    501000      0
         0     10000      0     -10000      0    125000];

L = zeros(6,6); L(2,6) = 1; L(3,5) = -1;
CRB = U * MRB * L

CRB =
      0      0      0      0      0      0
      0      0      0      0      0     1000
      0      0      0      0     -1000      0
      0      0      0      0      0     -1000
      0      0      0      0     10000      0
      0      0      0      0      0     10000
```

Note that the skew-symmetric property is destroyed by linearization. In other words, $\mathbf{C}_{RB}^* \neq -(\mathbf{C}_{RB}^*)^\top$

Summary: 6-DOF Rigid-Body Equations of Motion

Newton–Euler Equations of Motion about the CO

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

Rigid-body dynamics

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & -m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

Independent of the linear velocities u , v and w

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_b^b \end{bmatrix} = \begin{bmatrix} m & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m & 0 & -mz_g & 0 & mx_g \\ 0 & 0 & m & my_g & -mx_g & 0 \\ 0 & -mz_g & my_g & I_x & -I_{xy} & -I_{xz} \\ mz_g & 0 & -mx_g & -I_{yx} & I_y & -I_{yz} \\ -my_g & mx_g & 0 & -I_{zx} & -I_{zy} & I_z \end{bmatrix}$$

CG to CO Matrix Transformations

$$\mathbf{M}_{RB} := \mathbf{H}^\top(\mathbf{r}_{bg}^b)\mathbf{M}_{RB}^{CG}\mathbf{H}(\mathbf{r}_{bg}^b)$$

$$\mathbf{C}_{RB} := \mathbf{H}^\top(\mathbf{r}_{bg}^b)\mathbf{C}_{RB}^{CG}\mathbf{H}(\mathbf{r}_{bg}^b)$$

$$\mathbf{r}_{bg}^b = [x_g, y_g, z_g]^\top$$

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \mathbf{H}^\top(\mathbf{r}_{bg}^b) \begin{bmatrix} 0 & -mr & mq & 0 & 0 & 0 \\ mr & 0 & -mp & 0 & 0 & 0 \\ -mq & mp & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_z r & -I_y q \\ 0 & 0 & 0 & -I_z r & 0 & I_x p \\ 0 & 0 & 0 & I_y q & -I_x p & 0 \end{bmatrix} \mathbf{H}(\mathbf{r}_{bg}^b)$$

$$\mathbf{H}(\mathbf{r}_{bg}^b) := \begin{bmatrix} \mathbf{I}_3 & \mathbf{S}^\top(\mathbf{r}_{bg}^b) \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_3 \end{bmatrix}, \quad \mathbf{H}^\top(\mathbf{r}_{bg}^b) = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_3 \end{bmatrix}$$

摘要：6自由度刚体运动方程

关于质心的牛顿-欧拉运动方程

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

Rigid-body dynamics

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} m\mathbf{S}(\boldsymbol{\nu}_2) & -m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b)\mathbf{S}(\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{I}_b^b\boldsymbol{\nu}_2) \end{bmatrix}$$

Independent of the linear velocities u , v and w

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_3 & -m\mathbf{S}(\mathbf{r}_{bg}^b) \\ m\mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_b^b \end{bmatrix} = \begin{bmatrix} m & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m & 0 & -mz_g & 0 & mx_g \\ 0 & 0 & m & my_g & -mx_g & 0 \\ 0 & -mz_g & my_g & I_x & -I_{xy} & -I_{xz} \\ mz_g & 0 & -mx_g & -I_{yx} & I_y & -I_{yz} \\ -my_g & mx_g & 0 & -I_{zx} & -I_{zy} & I_z \end{bmatrix}$$

CG to CO Matrix Transformations

$$\mathbf{M}_{RB} := \mathbf{H}^\top(\mathbf{r}_{bg}^b)\mathbf{M}_{RB}^{CG}\mathbf{H}(\mathbf{r}_{bg}^b)$$

$$\mathbf{C}_{RB} := \mathbf{H}^\top(\mathbf{r}_{bg}^b)\mathbf{C}_{RB}^{CG}\mathbf{H}(\mathbf{r}_{bg}^b)$$

$$\mathbf{r}_{bg}^b = [x_g, y_g, z_g]^\top$$

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \mathbf{H}^\top(\mathbf{r}_{bg}^b) \begin{bmatrix} 0 & -mr & mq & 0 & 0 & 0 \\ mr & 0 & -mp & 0 & 0 & 0 \\ -mq & mp & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_z r & -I_y q \\ 0 & 0 & 0 & -I_z r & 0 & I_x p \\ 0 & 0 & 0 & I_y q & -I_x p & 0 \end{bmatrix} \mathbf{H}(\mathbf{r}_{bg}^b)$$

$$\mathbf{H}(\mathbf{r}_{bg}^b) := \begin{bmatrix} \mathbf{I}_3 & \mathbf{S}^\top(\mathbf{r}_{bg}^b) \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_3 \end{bmatrix}, \quad \mathbf{H}^\top(\mathbf{r}_{bg}^b) = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{S}(\mathbf{r}_{bg}^b) & \mathbf{I}_3 \end{bmatrix}$$

Chapter Goals - Revisited

- Understand that Newton's 2nd law and its generalization to the **Newton-Euler equations of motion is formulated in an "approximative" inertial frame** usually chosen as the tangent plane NED.
- Understand why we get a $\mathbf{C}_{RB}$ matrix when transforming the equations of motion to a BODY-fixed rotating reference frame instead of NED.
- Be able to write down and simulate the **rigid-body equation of motion about the CG** for a vehicle moving in 6 DOFs.
- Know how to **transform** the rigid-body equation of motion **to other reference points** such as the CO. This involves using the $\mathbf{H}$ -matrix (system transformation matrix) defined in Appendix C.
- Understand the:
 - 6 x 6 rigid-body matrix $\mathbf{M}_{RB}$
 - 6 x 6 Coriolis and centripetal matrix $\mathbf{C}_{RB}$ and how it is computed from $\mathbf{M}_{RB}$
 - Parallel-axes theorem and its application to moments of inertia

章节目标 - 再访

- 要理解牛顿第二定律及其推广到牛顿-欧拉运动方程，是在一个“近似的”惯性参考系中表述的，这个参考系通常选择为切平面NED。
- 理解为什么当将运动方程转换到机体固定的旋转参考系而不是NED时，我们会得到一个 $\mathbf{C}_{RB}$ 矩阵。
- 能够写出并模拟车辆在六自由度运动中关于质心的刚体运动方程。
- 了解如何将刚体运动方程转换到其他参考点，例如质心。这涉及使用附录C中定义的 $\mathbf{H}$ 矩阵（系统转换矩阵）。
- 理解：- 6 x 6 刚体矩阵 $\mathbf{M}_{RB}$ - 6 x 6 科里奥利和向心矩阵 $\mathbf{C}_{RB}$ 以及它如何从 $\mathbf{M}_{RB}$ 计算得出 - 平行轴定理及其在转动惯量中的应用